[Final Report Template]

[Block Y1B-2024-25]

Important Notes:

- Please, rename the file to FinalReport_<your_name>_<studentnumber>.**PDF** before uploading it.
- Upload this template to the 'Deliverables' folder in your BUas GitHub repository.
- You can add as many sections/ subsections as needed.
- You should structure sections based on paragraphs (you can add bullet points, numberings, and plots if needed).
- The final report should be submitted in **"PDF"** format only.

Introduction

You determine your business case by understanding the dataset and the features in it that you predict, classify or cluster based on machine learning models.

Problem Statement

A problem overview explaining your implemented baselines to examine problems that can be addressed by regression or classification or clustering.

This project focuses on analyzing and modeling a healthcare dataset to address key challenges and gain meaningful insights using machine learning techniques. The analysis incorporates both supervised and unsupervised learning methods to explore three distinct problem types: regression, classification, and clustering. Each problem type was approached with specific goals and carefully chosen models to solve the associated tasks.

For regression, the goal was to predict continuous outcomes, specifically the length of hospital stays represented by the variable time_in_hospital. This involved identifying numerical and categorical features that were likely to influence the target variable. Two regression models were implemented. The first was a Linear Regression model, which served as a baseline. Linear Regression was chosen for its simplicity and interpretability, making it easier to understand the relationships between features and the target variable. Features such as num_medications, num_lab_procedures, and discharge_disposition_id were selected because they directly reflect the complexity of the patient's medical treatment. The second model was a Random Forest Regressor, which was implemented to handle potential nonlinear relationships in the data. Unlike Linear Regression, Random Forest can

capture more complex patterns and also provides insights into feature importance, offering a deeper understanding of which variables have the most significant impact on hospital stay length.

For classification, the objective was to categorize patients based on specific outcomes, such as whether or not they were readmitted to the hospital. This task involved both numerical and categorical features that could influence readmission likelihood. A RandomForestClassifier was used for this problem, as it is a robust model capable of handling mixed data types and class imbalances. The model's performance was evaluated using the F1-score, a metric that considers both precision and recall, making it well-suited for imbalanced datasets. Features like admission_type_id, discharge_disposition_id, and num_medications were used to train the model because they provided meaningful information about patient admissions and treatment complexity, which are critical factors in determining readmission risk.

For clustering, the goal was to group patients into distinct clusters based on patterns in their medical data. This unsupervised learning task aimed to uncover underlying structures in the dataset, such as commonalities in laboratory procedures or medication usage. KMeans Clustering was applied as the primary algorithm, grouping patients based on similarities in selected numerical features. Standardized features, such as num_medications, num_lab_procedures, and number_inpatient, were used to ensure all variables contributed equally to the clustering process. The performance of the clustering model was evaluated using the Silhouette Score, which measures how well-separated and cohesive the clusters are. While the clustering results were not as strong as expected, the process provided valuable insights into the dataset's distribution and potential areas for further refinement.

Overall, this project involved a comprehensive application of machine learning techniques to address regression, classification, and clustering problems. Each model was carefully selected and tailored to the specific goals of the task, ensuring meaningful results and actionable insights from the data.

1) Exploratory Data Analysis (EDA)

The first step in EDA is to gain a general understanding of the dataset. For instance, to determine the total number of records in a specific column, such as encounter_id, we can write a query to calculate the total rows in that column. This helps us understand the overall size of the data and check for its completeness, ensuring that no key information is missing at the foundational level.

Next, we examined the distribution of different admission types. By analyzing this information, we gained insights into the variety of reasons patients were admitted to the hospital, which is crucial for identifying patterns in patient demographics or medical needs.

Another important aspect of the analysis involved exploring discharge dispositions. Using a query, we calculated how often each discharge disposition appeared in the dataset. The results were grouped by discharge type, sorted from most to least frequent, and the top 10 results were displayed. This analysis provided a clearer picture of where patients typically went after being discharged, such as home, nursing facilities, or other medical institutions.

To ensure data integrity, it was necessary to identify missing or anomalous data. For example, we checked the race column for missing values and calculated the occurrence of each value in the column. This helped us identify whether any significant gaps existed in the data and how evenly the data was distributed among different categories. Similarly, we investigated the weight column for missing or unusual values. In SQL, missing values are typically represented as NULL. When we checked for null values in the weight column, the result was zero, indicating that the column had no missing values.

The next step was to calculate the age distribution within the dataset. By querying the minimum and maximum values in the age column, we established the overall age range of patients in the dataset. We also calculated the number of patients in each age group to better understand the demographic breakdown. This analysis provided critical insights into the age groups most commonly represented in hospital admissions, which could influence further modeling decisions.

We then analyzed the contribution of different admission sources to hospital admissions. This task highlighted which sources—such as emergency rooms, physician referrals, or transfers from other facilities—were most commonly associated with patient admissions. Understanding this information is important for identifying trends in how patients enter the hospital system.

Another key analysis was determining the average time spent in the hospital for each admission type. Using SQL, we grouped the data by admission type and calculated the average length of hospital stay. The results were then sorted to display admission types with the longest stays first

2) Data Processing

Handling missing values was an important step in preparing the dataset. Missing values can lead to inaccurate models or biased predictions, so it was crucial to address them early. In my analysis, I found that some columns had a large proportion of missing values—more than 30-40%—and decided to drop those columns. This was because filling in so many missing values could add noise and reduce the reliability of the model. For numerical features, I chose to replace missing values with the mean. This method was simple and ensured that the central tendency of the data remained intact without removing any rows. For categorical features, I filled the missing values with the most frequent category. This approach helped maintain the structure of the dataset and avoided introducing unnecessary randomness.

Metadata was also an essential part of the process. Metadata acts as a guide to understanding what each column represents and how it should be treated. For example, discharge_disposition_id was an important feature that represents patient outcomes and could influence predictions like the length of hospital stays or readmission likelihood. By examining the metadata, I could differentiate between numerical and categorical features and decide on appropriate preprocessing methods, such as encoding categorical data or scaling numerical data. Without this understanding, I might have misinterpreted certain columns, which could have affected the overall analysis.

Visualizations played a key role in understanding the dataset and making preprocessing decisions. For instance, I used distribution plots to check the spread of numerical features like num_medications and time_in_hospital. These plots helped me spot skewed data or potential outliers that might need to be handled. A correlation heatmap was another valuable tool, as it showed how strongly numerical features were related to one another. This helped me identify and address multicollinearity, which can confuse models if not managed. Box plots were helpful in visualizing outliers and understanding the range of values in numerical features, while scatter plots allowed me to explore relationships between different variables, such as num lab procedures and time_in_hospital.

To address missing values, I created a heatmap that visually highlighted where they were located in the dataset. This made it easier to identify columns or rows with significant missing data and decide whether to drop or fill them. Additionally, I used a feature importance plot from the Random Forest model to understand which features were the most relevant to the predictions. This was crucial in focusing on the variables that truly mattered and avoiding unnecessary complexity in the models.

Overall, the data processing steps ensured that the dataset was clean and ready for modeling. Addressing missing values, understanding metadata, and using visualizations allowed me to make informed decisions about which features to keep and how to prepare them. This helped improve the quality and reliability of the models I developed later.

3) Machine Learning Models

A model overview explaining the process of creating and training and evaluating the implemented regression, classification and clustering models and their iterations based on feature selections and fine tuning the parameters

- Regression Models

For the regression task, I focused on predicting the length of hospital stays, represented by the time_in_hospital variable. To achieve this, I carefully selected features that were logically and statistically connected to the target variable. The process of choosing these features involved looking at their relationship with the target and their potential to provide meaningful insights into the problem.

One of the features I selected was num_medications, which refers to the total number of medications prescribed to a patient during their hospital stay. This feature seemed very relevant because patients with more severe conditions or complications typically require more medications. For example, someone admitted for a minor issue may only need one or two medications, while a patient with a chronic or complex illness may require a much larger number. Since the number of medications can reflect the overall complexity of the patient's medical needs, it made sense that this feature could be a strong predictor of how long a patient would stay in the hospital.

Another feature I included was num_lab_procedures, which captures the total number of laboratory tests a patient underwent. This feature also seemed crucial because it provides insight into the level of investigation or monitoring required for a patient. Patients with more lab procedures are often being monitored for complex or critical conditions, which may lead to extended hospital stays. For example, a patient with frequent blood tests, imaging scans, or other diagnostic procedures is likely to have a longer stay than one with only basic tests.

I also chose discharge_disposition_id, which encodes the way the patient was discharged from the hospital. This feature includes categories such as discharged to home, transferred to another care facility, or expired. This feature was particularly important because it reflects the outcome and severity of the patient's condition. For instance, a patient who is discharged to a rehabilitation center might have had a longer hospital stay due to the need for ongoing care, while someone discharged home quickly likely had a simpler case. Including this feature helped capture the broader context of the patient's condition and their pathway after leaving the hospital.

The selection of these features wasn't random; it was guided by medical logic and patterns I observed in the data. For example, I used scatter plots to visualize the relationships between these features and time_in_hospital, and I noticed trends that supported their inclusion. Additionally, when I trained the models, these features consistently contributed to better predictions, which reinforced their importance.

Overall, the features I selected for regression were chosen because they directly or indirectly reflected the factors influencing hospital stay length. By focusing on features that are both meaningful and relevant, I aimed to create a model that could make accurate and interpretable predictions.

After preprocessing the data, I trained the Linear Regression model using the least squares method and evaluated its performance using metrics like Mean Absolute Error (MAE) and Mean Squared Error (MSE). This helped me understand how well the model captured the relationships in the data.

I also implemented a Random Forest Regressor to see if a more complex, nonlinear model could perform better. Random Forest is known for its ability to handle complex patterns and relationships, and it also provides insights into feature importance. After training the Random Forest model, I compared its MAE and MSE with those of the Linear Regression model. While Random Forest showed a slightly higher error, it provided additional insights into which features were most important for predicting time_in_hospital. This comparison allowed me to weigh the benefits of simplicity and interpretability in Linear Regression against the flexibility and complexity of Random Forest.

- Classification Models

The classification task focused on predicting categorical outcomes, such as whether a patient would be readmitted to the hospital. For this task, I used a RandomForestClassifier due to its ability to handle mixed data types, its robustness against overfitting, and its capacity to work well with imbalanced datasets.

Feature selection for classification was crucial. I analyzed the dataset to identify features that were likely to influence the outcome. Features such as admission_type_id, discharge_disposition_id, and age were included because they are directly related to the patient's admission and treatment process. Additionally, features like num_medications and number_inpatient were included because they provide insight into the severity of the patient's condition, which could impact the likelihood of readmission.

Since classification models require numerical input, I needed to convert categorical variables into numerical form. For this, I used one-hot encoding for features like admission_type_id and discharge_disposition_id, which had a manageable number of categories. One-hot encoding creates binary columns for each category, allowing the model to understand these features without introducing ordinal relationships that don't exist.

To evaluate the model, I used metrics like accuracy, precision, recall, and the F1-score. The F1-score was particularly important because it balances precision and recall, making it suitable for imbalanced datasets where one class might dominate. During training, I noticed that the model performed well in capturing patterns, but to further optimize its performance, I focused on hyperparameter tuning.

- Clustering Models

The clustering task aimed to group patients based on similar patterns in the dataset, without any predefined target variable. This unsupervised learning approach was particularly useful for

uncovering hidden patterns or natural groupings, such as identifying clusters of patients with similar treatments or medical conditions.

To prepare the dataset for clustering, I focused on numerical features that were relevant and standardized them to ensure fair contribution during the clustering process. Features like num_medications, num_lab_procedures, and number_inpatient were particularly important because they reflected the intensity and complexity of the care each patient received. For instance, patients with high values in these features might represent those with more severe conditions, making these attributes key in determining clusters.

Before applying the clustering algorithm, I used scaling because KMeans relies on distance metrics like Euclidean distance, which are sensitive to feature magnitudes. Without scaling, features with larger ranges could dominate the clustering results, leading to biased groupings. To address this, I standardized all numerical features using a StandardScaler, ensuring that each feature had a mean of 0 and a standard deviation of 1.

After preprocessing, I applied the KMeans algorithm. Choosing the number of clusters was challenging, as this parameter greatly impacts the results. I experimented with different values for the number of clusters (k), ranging from 2 to 10, and evaluated the results using the Silhouette Score. This score helped me assess how well-separated and cohesive the clusters were. However, I faced significant challenges when trying to optimize the clustering process using GridSearchCV. Specifically, I encountered an issue where I could not calculate the Silhouette Score during the grid search, as the scoring function returned NaN values. To troubleshoot, I tested different datasets, including the Iris dataset, to determine whether the problem was specific to my data. Unfortunately, I experienced the same issue even with the Iris dataset, which confirmed that the problem was not solely related to the original dataset.

To overcome this limitation, I created a custom function to calculate the Silhouette Score and integrate it into the process. This allowed me to obtain the results I needed and continue with the clustering evaluation. Through further research, I discovered that using GridSearchCV for clustering tasks can sometimes be problematic due to the lack of predefined labels and the differences in how unsupervised models are evaluated compared to supervised ones. This limitation helped me better understand the challenges of clustering optimization and adapt my approach to address them.

Unfortunately, the Silhouette Score for all tested values of k remained relatively low, suggesting that the data did not form clear, well-defined clusters. While the clustering results were not as strong as I had hoped, this process provided valuable insights into the dataset's structure and potential areas for improvement. For instance, refining feature selection, experimenting with alternative clustering algorithms like DBSCAN or Gaussian Mixture Models, or using more specialized metrics for clustering could lead to better results in the future.

- Model Selection

Model selection and hyperparameter tuning played a critical role in improving the performance of all models. For both classification and regression tasks, I used GridSearchCV, a powerful tool for

systematically testing combinations of hyperparameters to find the optimal configuration for the model.

For the classification task with RandomForestClassifier, I tuned parameters such as the number of trees (n_estimators), the maximum depth of each tree (max_depth), and the minimum number of samples required to split a node (min_samples_split). Using GridSearchCV, I specified a range of values for each parameter and allowed the tool to perform cross-validation on each combination. Cross-validation was essential here because it ensured the model's performance was evaluated on multiple subsets of the data, reducing the likelihood of overfitting or underfitting.

Similarly, for the regression task with Random Forest, I tuned hyperparameters like the number of trees and the maximum features considered at each split. GridSearchCV provided insights into which combination of parameters yielded the lowest Mean Squared Error (MSE) on the validation set.

Feature engineering was also an important part of the process. For example, I transformed categorical data into numerical form using encoding techniques and scaled numerical features to ensure consistency across models. Additionally, I experimented with dropping certain features that had little correlation with the target variable, as they could add unnecessary noise.

Ultimately, model selection was based on the performance metrics and interpretability. For regression, Linear Regression was chosen because it had the lowest MSE and was easier to interpret than the more complex Random Forest. For classification, the RandomForestClassifier was selected due to its robust performance and ability to handle imbalanced data effectively. While KMeans did not yield strong results for clustering, the insights gained from the process highlighted areas for future exploration.

Overall, the iterative process of selecting features, tuning parameters, and evaluating performance ensured that the best models were chosen for each task. This approach not only improved accuracy but also provided a deeper understanding of the dataset and the factors influencing the outcomes.

not satisfactory, so I decided not to use it as a final model. Instead, I concluded that clustering might require a more advanced algorithm or additional preprocessing steps to achieve meaningful results.

Overall, the modeling process involved carefully selecting features, fine-tuning parameters, and comparing the performance of different models. This iterative process helped ensure that the best model was chosen for each task, balancing accuracy, interpretability, and the specific requirements of the problem.

Analyze and describe the reasons for selecting one of your implemented models as a best option for each task.

4) Conclusion and Reflection

This project demonstrated the importance of tailoring models to specific tasks and iteratively refining them through feature selection, preprocessing, and hyperparameter tuning. The regression and classification models performed well and provided actionable insights, while the clustering task revealed opportunities for further exploration.

Looking back, one of the most valuable lessons was the importance of understanding the dataset before applying any models. Properly handling missing values, scaling features, and converting categorical data were all critical steps that directly impacted model performance. Another key takeaway was the role of evaluation metrics in guiding decision-making. For instance, using MSE and MAE for regression, the F1-score for classification, and the Silhouette Score for clustering ensured that the models were assessed fairly and accurately.

5) References (if there is any)

[References]